Project Venom

Scientific dossier for feasibility assessment

David

Longitudinal observation of the transition towards loss of viability in independent tumour samples.

1000 → 1 → 0

Itinerant Virus Model (MVI)

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Proposal and methodology for specialist discussion.
Translation of the Spanish version dated 8 October 2026.
No experimental results generated by this project.

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1. Summary and requested decision

The project proposes exploring whether monitoring tumour samples over time as they lose viability can reveal a recurrent biological candidate that is difficult to distinguish while viable cellular refuges remain available. The hypothesis concerning cancer origins attributes to Venom—a virus or similar biological entity—an initiating role shared across different cancers and the capacity to remain in or leave the environment it transforms.

The immediate question is narrower: **can a candidate be identified during this transition with concordant identity, recurrence across donors, and a temporal and comparative relationship that warrants investigation?** Observation extends from viable tissue to an operational zero and the subsequent period, examining both the cellular compartment and the associated medium.

An assessment of feasibility and informational value is requested before committing to a broad search. The dossier organises the assumptions, comparators, possible outcomes and decisions needed to turn the proposal into a study. The responsible team must resolve the technical design, thresholds and sample size.

Proposed development pathway. Feasibility assessment → bounded exploration → definition of a candidate and its rules → confirmation in independent samples → subsequent causal assessment, if appropriate. Each stage has its own decision to proceed or stop.

Criterion for further interest. A signal warrants further work if it can be characterised, distinguished from ordinary sources of error, and observed again under predefined rules. The initial contribution would be to identify an opportunity for observation and, potentially, a candidate. Attribution of a common origin would require additional work on causality and scope.

Request to the reviewer. Determine whether there is a technically accessible window, whether the comparators can distinguish relevant explanations, and whether the expected information justifies the cost of a bounded pilot. Three recommendations are possible: develop a pilot; reformulate before starting; or do not proceed, explaining the decisive limitation.

2. General hypothesis and testable version

General hypothesis concerning origins. Venom could initiate transformations in different tissues. Altered cells and their descendants could retain consequences of that intervention even when the initiator is no longer detectable. The proposal retains its viral orientation and its ambition to investigate a shared root cause.

Working experimental hypothesis. In a specifiable fraction of samples from a defined population, a candidate present during monitoring depends on viable cellular refuges for the postulated concealed state. Reduction of those refuges is associated with a transition observable using a system whose scope is declared.

This version depends on four separate conditions: presence during the window studied; dependence on the defined refuge; a transition that generates observable information; and instrumental capacity to record it. The broad hypothesis alone neither specifies how frequently these conditions occur nor guarantees that they are met.

Testable expectation. After exploration, an identity rule, a window relative to viability, a minimum recurrence of interest, and a contrast against comparators must be fixed. Confirmation will assess this combination in new data. “Something unusual appears” is an exploratory observation, not the definition of a confirmatory result.

Main alternatives. Ordinary cellular deterioration; host structures or material; known viruses; contamination; instrumental or analytical artefacts; a secondary tumour component; and changes caused by ex vivo conditions. The design must specify which comparison informs each alternative.

Causal scope. Following a sample that is already tumorous does not, by itself, reconstruct disease initiation. A real, recurrent candidate temporally associated with deterioration may be a cause, an accompanying factor or a consequence. Its discovery would justify characterisation and further questions, while that attribution remains open.

Extensions concerning intergenerational transmission, immunity, action on other biological entities and pharmacological signatures fall outside the initial test. They remain in the programme as distinct hypotheses, without retrospectively modifying the commitments of the cellular version.

3. Rationale and relationship to previous work

There are experimental precedents for separating an initial intervention from the maintenance of a transformation. Shen, Zhu and Shenk studied a transformation system in rat cells involving transient viral activity [D1]. Its relevance here is conceptual: distinguishing initiation from persistence. The specific Venom proposal requires its own evidence.

Zapatka and colleagues investigated viral associations in pan-cancer data through genomic and transcriptomic analyses, incorporating assessments of contamination and batch effects [D2]. This precedent places the search within an existing field and makes documenting the origin of any common signal particularly important.

The proposed contribution to be assessed is a temporal window: observing declining viability and relating it to changes in detectability. Its novelty relative to other designs must be examined through specialist review; this dossier establishes neither priority nor bibliographic completeness.

Deterioration offers an opportunity for observation, but also generates biological material that complicates interpretation. Atkin-Smith and colleagues described vesicle formation during apoptosis [D3]. Bullock and colleagues analysed difficulties distinguishing subcellular structures from viral particles by electron microscopy [D4]. Both studies support the need for comparisons and identity verification.

Models of transformation through defined alterations, such as that described by Drost and colleagues in human organoids [D5], are relevant to assessing whether a pattern depends on the tumour state or on a particular intervention. Their equivalence to naturally occurring tumour tissue must be justified for each comparison.

MISEV2023 provides recommendations for characterising extracellular vesicles and documenting study conditions [D6]. It is a methodological reference for distinguishing properties, provenance and measurement; it is not adopted as a protocol for identifying Venom.

Rationale for assessment. There is a specific temporal question and a recognisable set of alternatives. The main uncertainty is whether the proposed window allows observation that is sufficiently specific and reproducible to add information beyond ordinary deterioration. This is the first issue the feasibility review should resolve.

4. Operational definitions and observation limits

Viable cellular refuge. A cell that meets the agreed viability criteria and, within the version being assessed, could sustain the proposed concealed state. “Possible refuge” does not imply occupancy by the agent.

Conceptual count. 1000 viable cells = 1000 possible hiding places; 1 = 1; 0 = none. This describes exhaustion of that class of refuge. The number 1000 does not determine the number of donors, samples or experimental units.

Operational zero. Absence of detectable viability within a declared volume, coverage and sensitivity limits. The team must define convergent indicators and how discrepancies will be resolved. Heterogeneous tissues may retain viable niches outside the observed field; declaring zero requires addressing this coverage.

Transition. The sequence from initial viability through deterioration, the last detectable cells, operational zero and subsequent monitoring. The current hypothesis does not require physical exit or an event occurring exactly when the last cell dies.

Longitudinal observation. A temporal record with documented continuity and gaps. Destructive measurements or measurements incompatible with continued monitoring are not equivalent to repeatedly observing the same intact cell. Inferences across fragments or related samples must be identified as such.

Candidate signal. A change or component that meets a documented exploratory rule and warrants characterisation. **Concordant identity** means convergence of relevant properties attributable to the same object; visual resemblance or an isolated match is insufficient to group observations under a single identity.

Recurrence. Reappearance under comparable rules in independent biological units. **Differential pattern** is the predefined relationship against comparators, which may involve presence, frequency, magnitude or temporal relationship. Its form is determined before the confirmatory phase.

Instrumental scope must be delineated. The historical 20–200 nm filter is withdrawn as an a priori definition of Venom. Broadening the search requires declaring blind spots and measurement compatibility, rather than assuming that a combination of techniques observes every possible entity.

5. Conceptual design and study units

Primary biological unit. The donor. Fragments, fields and repeated observations are nested within their source. They may improve precision for a sample, but do not increase the number of independent donors. Any analysis must preserve this structure.

Population of interest. A selection of tumour types defined by the question and the possibility of interpretable monitoring. Pan-cancer diversity is a programme objective; a pilot should start with a justifiable scope rather than indiscriminately assembling many types of uneven quality.

Required information. Sample provenance and classification; relevant prior treatment; preanalytical interval and conditions; initial state; tissue composition; information on known infections where relevant; batches, instruments and processing. Factors that may confound signal and viability must be recorded under agreed criteria.

Monitoring of each sample. Preserve its identity and trajectory, record viability and signal within a common temporal framework, examine the interior and associated medium, and document discontinuities. Compare both elapsed time and biological state: two samples with the same time ex vivo may have different viabilities.

Physical independence. The book's "independent chambers" translate into sample separation and traceability of contacts, materials and processing. Independence is also assessed with respect to prior contamination, reagents, batches, operators and shared analytical decisions.

Bias reduction. Define selection and exclusions; balance technical factors between groups where feasible; blind the analyst to the source category where the task permits; retain raw data, metadata and negative results. The anomaly criterion must prevent the observer's expectations from determining what is recorded.

Separation of phases. The exploratory phase generates rules and candidates. The confirmatory phase retains fixed rules and uses new samples. Revising the rules reopens exploration; this must be recorded and not presented as independent confirmation.

This framework does not prescribe obtaining, culturing, propagating or manipulating an agent. The first technical deliverable requested from the team is a measurement and comparison plan that allows the feasibility of observation to be judged.

6. Proposed methodology for developing the study

The methodology is organised into work packages with verifiable outputs. This version specifies what must be resolved and how measurements should be related. The institution's standard operating procedures must establish execution parameters once samples, equipment and safety have been assessed.

M1. Selection and traceability. Define the population, inclusion and exclusion criteria, and comparators. Assign identifiers linking donor, sample, fragment, compartment and record. Histopathological review must establish composition and representativeness. Deliverable: a catalogue of units and a complete preanalytical record, preserving confidentiality.

M2. Viability assessment. Select complementary measures of cellular state, such as membrane integrity, metabolic activity and morphological features, justifying each contribution. A single signal may vary with the type of death or measurement conditions. Define convergence, discrepancies, spatial coverage and limits for declaring operational zero. Deliverable: a viability rule with its uncertainties.

M3. Temporal monitoring. Assess longitudinal imaging modalities compatible with the material, including minimally perturbing optical observation where appropriate. Define spatial scope, temporal resolution, synchronisation and recording of interruptions. Monitoring must cover the entire transition; its relationship to zero is analysed from the record. Deliverable: viability and signal trajectories with comparable timestamps.

M4. Complementary characterisation. When a candidate signal is observed, assess the information that available structural and molecular characterisation could provide: higher-resolution imaging, nucleic-acid or protein profiles, according to the candidate type and authorised scope. Techniques are selected to answer an identity question, not to accumulate modalities. Deliverable: evidence enabling properties to be attributed to the same candidate and distinguishing host structures or known agents [D4, D6].

M5. Relationships between compartments and measurements. Link intracellular observation to the associated medium and complementary analyses. If a measurement consumes material, specify whether it concerns the same object, another fraction or a related sample. Deliverable: a correspondence map showing where monitoring is direct and where inference is made across units.

M6. Quality and interpretation. Incorporate controls and references, blinded analytical review where possible, batch review and agreed exclusion criteria. Preserve the complete record. Deliverable: an evaluable dataset, incidents and justified decisions. The analysis section defines how to derive results from that dataset.

7. System development and parameters to be finalised

First stage: feasibility of the observation system. Verify that a viability trajectory can be followed, its coverage documented, and a signal linked to the sample. Analytical references and institutionally authorised materials allow specific capabilities to be assessed before attributing findings to an unknown candidate.

Second stage: bounded exploration. Apply a prospective recording plan to a bounded set of samples and comparators selected for relevance and availability. Describe signals, failures and sources of variation. The output is a feasibility assessment and, if appropriate, a provisional candidate definition.

Third stage: confirmation. Register the population, primary outcome, identity rules, temporal window, comparators, sample size and analysis before examining new data. Confirmation must allow its expectations to fail; its rules are not adjusted to accommodate every observed result.

Material parameters. Admissible sample types, representativeness, prior-treatment factors, acceptable preanalytical intervals and minimum initial state. Proposed responsibility: the sample and pathology team, together with the experimental team.

Measurement parameters. Observable sample volume and fraction; viability indicators; acquisition resolution and frequency; duration of monitoring after zero; quality thresholds; criteria for matching signal and identity. Proposed responsibility: cell biology and analytical platforms, with statistical support.

Inference parameters. Candidate rule, minimum detectable frequency of interest, main contrast, management of multiplicity, allowable losses and desired precision. Proposed responsibility: the scientific and biostatistics team. These must be established before sizing the confirmatory study.

Stopping conditions. Loss of traceability, unresolved contamination, insufficient coverage of the relevant period, incompatible measurements or failure to meet institutional conditions. Each incident is recorded as a failure of that unit or of the design, according to its scope.

Document required before execution. An approved protocol specifying objectives, institutional procedures, responsibilities, controls, data to collect, quality criteria, analysis plan and decision rules. This methodology provides its working basis; completing the parameters requires system-development tests and specialist assessment. Procedures for isolating, amplifying or propagating an unknown agent are not included here.

8. Comparators and the role of controls

Comparable non-tumour tissue. Provides information on host deterioration and allows differences associated with tumour status to be studied. Its origin, composition and viability trajectory must support interpretation of the comparison. Adjacent tissue is not automatically equivalent to independent healthy tissue.

The MVI allows presence in apparently normal cells. Therefore, negativity in all non-tumour tissue cannot be imposed as an assumption. The comparison requires a specified differential expectation. If the same pattern is observed in both groups, it will be necessary to assess whether the proposed association is lost or a different question is being studied.

Transformation through a known mechanism. A tumour model with defined alterations allows assessment of whether that state is sufficient to produce the pattern [D5]. Its history and differences from naturally occurring tissue affect inference. A specific biological comparison is being evaluated, without declaring by fiat that every unknown entity is absent.

Technical and processing blanks. Provide information on signals originating in materials, the analytical environment or processing. A relationship between the signal and a batch or shared pathway must be examined before attributing biological recurrence.

Analytical references. The system's capacity to record relevant classes of signal must be assessed using appropriate, safe references. Because the candidate is unknown, a reference validates specific system capabilities; it does not by itself determine sensitivity for Venom.

Comparable states. Tumour samples and comparators must be interpreted in relation to viability, composition, quantity of material and recording quality. Equal monitoring time does not ensure equivalent deterioration. A signal difference may be due to one of these variables.

Implication for universality. If a universal requirement for the agent is proposed, models of transformation through defined pathways raise a question of scope that must be addressed. The initial test must specify its target population, keeping the broad etiological ambition separate from the immediate comparison.

Before starting, each control will have a question, an expectation and an interpretative consequence. This correspondence is more useful than a long list of controls without a defined function.

9. Outcomes, variables and analysis strategy

Primary outcome of the feasibility pilot. Proportion of units with interpretable monitoring under prior criteria for coverage, viability-assessment quality and linkage between signal and sample. This allows a decision on whether the system addresses the temporal question. Its acceptance threshold remains to be agreed.

Primary outcome of confirmation. Proportion of evaluable donors meeting a predefined rule for candidate, identity and temporal relationship, accompanied by the specified contrast against comparators. The pilot should help select a primary outcome; an extensive signal search must not retrospectively become multiple primary outcomes.

Secondary variables. Temporal distribution relative to viability; concordance across observation modalities; recurrence by tumour type; relationship with composition and preanalytical factors; associations with batches and other technical sources. Subgroup analyses must be identified as planned or exploratory.

Unit of inference. Retain the donor as the biological unit and model the dependence of repeated observations or summarise them under an agreed rule. Preserve matching when paired comparisons are used. The exact analytical model depends on the variable and the data distribution that the pilot makes available.

Multiplicity and selection. The number of signals, windows, techniques and comparisons creates opportunities for chance matches. Candidate generation must be separated from confirmation, and a strategy for controlling false findings appropriate to the phase must be defined. Rejected candidates and the reasons for rejection must also be retained.

Missing data. Distinguish loss to monitoring, contamination, failed measurement and biological exclusion. Record whether loss is associated with deterioration or sample type. Report the numbers received, evaluable and excluded; excluding cases does not automatically make the remainder representative.

Presentation. Individual trajectories, donor counts, uncertainty in estimates, comparisons by group and breakdown by batch. Representative images must accompany, rather than replace, the full record and the criteria for their selection.

10. Sample size and the meaning of absence

Study size depends on the minimum frequency of interest, the evaluable fraction, achievable sensitivity, the chosen comparison and heterogeneity between tumours. None of these parameters can be estimated from the count 1000 → 1 → 0.

For planning, the team must separate the probability that the candidate is present from the probability of detecting it when present. An observed frequency combines both, together with opportunities for error. An infrequent agent and a poorly detected agent can yield few positives for different reasons.

Mathematical illustration, not the study's sample-size calculation. If n donors were independent and comparable, detection were perfect, and each had the same probability p of being positive, the probability of observing zero positives would be (1 – p)^n. With zero positives, the exact one-sided 95% upper bound for p would be 1 – 0.05^(1/n), approximately 3/n when n is sufficiently large.

These idealised conditions have not been established for the project. With unknown sensitivity, absence of detections constrains the detectable frequency under the system used, not biological prevalence directly. Batch clustering, dependence between samples and heterogeneity further reduce the usefulness of a simple binomial calculation.

Required prior decision. Define which detectable frequency and which size of difference would warrant attention. Then estimate the number of evaluable units and initial donors needed, incorporating expected losses and the comparison structure. The calculation applies to the delimited version, not to an unrestricted search.

Informative negative. Absence of the specified pattern with coverage and quality sufficient for the declared expectation. Independent repetition weakens that experimental version. **Non-evaluable result.** The window was not adequately observed or a necessary condition failed; this leads to design review without adding support for the hypothesis.

The qualification "the agent may have left" belongs to the general model. For the test to have value, the version assessed must commit to presence or detectability in a defined fraction; it cannot withdraw that commitment after obtaining the result.

11. Rules for interpretation and further work

Signal linked to blanks, batches or processing. Investigate and resolve the technical source. Biological attribution is suspended while that explanation remains open. Repetition of the same error is not independent recurrence.

Comparable pattern in deteriorating tumour and non-tumour tissue. Assess whether it reflects a general cellular process [D3, D4]. If the specified difference is absent, the comparison does not support that version. It may retain interest for a different, clearly identified question.

Recurrent candidate with concordant identity and a reproduced contrast. Justifies additional characterisation and replication outside the discovery set. The next decision concerns the candidate; a shared etiological scope is assessed later.

Promising signal with unresolved identity. Retain the category of exploratory observation. Resolve the linkage between properties of the same object before grouping findings from different tumours. An image and a molecular signal from the same sample could originate from different components.

Repeated informative absence. Revise or abandon the delimited version according to the expectations set. **Persistent measurement infeasibility.** Stop the current design or reformulate it before expanding the sample collection. The programme's value depends on obtaining interpretable answers.

Independent replication. Requires new biological units and an implementation that examines shared sources of error, ideally by another team. Reanalysing the same file or repeating fragments from the same donor does not replace this task.

Subsequent causal study. If a robust candidate exists, its temporal relationship with the initiation of transformation, its mechanistic role and its explanatory power against alternatives could be examined. This document does not prescribe experiments involving infectivity, propagation or disease induction.

Decisions will be documented using the complete set of results, including exclusions, negatives and version changes. Continuing, reformulating and closing are legitimate outcomes of a well-delimited assessment.

12. Feasibility, resources and questions for the reviewing team

Material and institutional framework. Determine sample availability and provenance, adequacy of consent, ethical review and data protection. The responsible institution must assess containment, exposure and material management. The document does not independently assign a biosafety level to an unknown candidate.

Measurement. Assess whether loss of viability can be followed with sufficient coverage and whether complementary modalities can link a temporal signal to an identity. Identify limits, interference, unobserved volume and effects of monitoring itself.

Expertise. Histopathology and sample handling, cell biology, virology, analytical characterisation and imaging, biostatistics and data management. Roles may be distributed among professionals; the criterion is to cover the decisions, not merely assemble a list of specialties.

Resources to estimate. Access to material; instrument time; characterisation work; controls; record storage; analysis; independent review and replication. A decision to expand requires a pilot-based estimate of costs and monitoring capacity. No amounts are budgeted without knowing the platform.

Proposed conditions for proceeding. An observable window; a defensible definition of zero; relevant comparators; a signal rule that can be finalised; sufficient traceability; and the possibility of an interpretable negative result. The corresponding quantitative thresholds must be fixed before the stage intended to provide confirmation.

Questions for assessment. Is the formulated dependence on refuge testable with this material? What temporal information would be gained relative to existing designs? Which alternative would be hardest to exclude? Can identity be related to time of appearance? What failure would make increasing the sample collection unnecessary? What result would justify independent replication?

Dossier conclusion. The proposal has a defined question and outstanding technical issues that allow a concrete assessment. The requested decision is whether a bounded pilot can provide enough information to justify its cost, or whether the question should be reformulated first. The project's interest lies in turning a hypothesis concerning origins into an examinable opportunity for observation.

13. Relationship between documents and version control

Narrative book-document. Explains the origin, analogies and evolution of the idea for a general reader. Its function is to make the project understandable.

Scientific dossier. Translates that idea into hypotheses, variables, comparisons and research decisions. It must be independently readable by a specialist. Its function is to enable an assessment of relevance and feasibility.

Master backup. Preserves blocks 0–32 and the historical documents in full. Records how the proposal was reached and also preserves formulations subsequently modified.

The current version retains: Venom as a virus or similar biological entity; the distinction between agent and transformed cell; longitudinal observation inside and outside; independent samples; and the count of viable cellular refuges.

The historical formulations of mandatory exit when the last cell dies and a 20–200 nm filter are replaced by monitoring of the transition and declared instrumental scope. Extensions to other biological supports, pharmacological signatures and transmission remain separate lines of inquiry.

This version 1.0 organises a proposal for assessment. It has not been reviewed or approved by an external scientific team. Contributions from such a review must be recorded in a new version, identifying changes to hypothesis, population, measurement, comparators or interpretation criteria.

Conceptual closing shared with the book: Let us be watching when the last one disappears.

14. Selected references

A targeted selection addressing the rationale and design issues; this is not a systematic review. References consulted and checked in the Spanish source version: 8 October 2026. The cited work supports specific statements, not an endorsement of the project by its authors. This translation does not constitute a new reference check.

D1. Shen Y, Zhu H, Shenk T. Human cytomegalovirus IE1 and IE2 proteins are mutagenic and mediate “hit-and-run” oncogenic transformation in cooperation with the adenovirus E1A proteins. Proceedings of the National Academy of Sciences. 1997;94:3341–3345.

https://doi.org/10.1073/pnas.94.7.3341

Experimental precedent for transient viral intervention and transformation in rat cells.

D2. Zapatka M et al. The landscape of viral associations in human cancers. Nature Genetics. 2020;52:320–330.

https://doi.org/10.1038/s41588-019-0558-9

Pan-cancer search for viral associations; a reference for contamination, batch effects and the scope of previous searches. The updated publisher version was consulted for the source dossier.

D3. Atkin-Smith GK et al. A novel mechanism of generating extracellular vesicles during apoptosis via a beads-on-a-string membrane structure. Nature Communications. 2015;6:7439.

https://doi.org/10.1038/ncomms8439

Vesicle formation during apoptosis as a relevant biological alternative.

D4. Bullock HA, Goldsmith CS, Zaki SR, Martines RB, Miller SE. Difficulties in Differentiating Coronaviruses from Subcellular Structures in Human Tissues by Electron Microscopy. Emerging Infectious Diseases. 2021;27:1023–1031.

https://doi.org/10.3201/eid2704.204337

Problems in the morphological identification of particles and cellular structures.

D5. Drost J et al. Sequential cancer mutations in cultured human intestinal stem cells. Nature. 2015;521:43–47.

https://doi.org/10.1038/nature14415

A transformation model with defined alterations, relevant to discussion of comparators and causal scope.

D6. Welsh JA et al. Minimal information for studies of extracellular vesicles (MISEV2023): From basic to advanced approaches. Journal of Extracellular Vesicles. 2024;13:e12404.

https://doi.org/10.1002/jev2.12404

Recommendations for extracellular-vesicle characterisation and documentation.